QUALIFICATION REQUIREMENTS REMOVED

MIL-M-38510/43B
28 February 1984
SUPERSEDING
MIL-M-38510/43A
24 May 1982

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, LOW POWER, PRIORITY ENCODERS, MONOLITHIC SILICON

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS REVISION.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon TTL, low power, priority encoders. One product assurance class and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, with the exception that the "JAN" or "J" certification shall not be used.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	8-input priority encoder

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outlines, (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package

1.3 Absolute maximum ratings.

Storage temperature	-65°C to +150°C
Temperature (ambient) under bias	-55°C to +125°C
V_{CC} pin potential to ground pin	-0.5 V to +7.0 V
*Input voltage (dc)	-0.5 V to +5.5 V
*Input current (dc)	-30 mA to +5.0 mA
Voltage applied to outputs (output high)	-0.5 V to +V_{CC} value
Output current (dc) (output low)	+30 mA
Thermal resistance, junction-to-case (θ_{JC})	
Case F	0.09°C/mW for flat package
Case E	0.08°C/mW for dual-in-line package

*Either input voltage limit or input current limit is sufficient to protect the inputs.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5962

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1.4 Recommended operating conditions.

Supply voltage	4.5 V minimum to 5.5 V maximum
Minimum high level input voltage	2.0 V
Maximum low level input voltage	0.7 V
Case operating temperature range	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specifictaion to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification For.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and 1.2.3 herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Truth tables. The truth tables shall be as specified on figure 2.

3.2.3 Logic diagram. The logic diagram shall be as specified on figure 3.

3.2.4 Case outline. The case outline shall be as specified in MIL-M-38510 and in 1.2.3 herein.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. Electrical test requirements shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510. The "JAN" or "J" certification mark shall not be used.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/	Device type	Limits Min	Limits Max	Unit
High level output voltage	V_{OH}	V_{CC} = 4.5 V, I_{OH} = -0.4 mA	01	2.4	---	V
Low level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 4.8 mA	01	---	0.3	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -10 mA	01	---	-1.5	V
High level input current at input $\overline{0}$	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V	01	---	20	µA
High level input current at all other inputs	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 2.4 V	01	---	40	µA
High level input current at input $\overline{0}$	I_{IH3}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	01	---	100	µA
High level input current at all other inputs	I_{IH4}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	01	---	200	µA
Low level input current at input $\overline{0}$	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.3 V	01	-176	-400	µA
Low level input current at all other inputs	I_{IL2}	V_{CC} = 5.5 V, V_{IN} = 0.3 V	01	-352	-800	µA
Short circuit output current	I_{OS}	V_{CC} = 5.5 V 2/	01	-2.5	-25	mA
Supply current	I_{CC}	V_{CC} = 5.5 V	01	---	22	mA
Propagation delay time, low-to-high output from data to $\overline{EO}$	t_{PLH1}	V_{CC} = 5.0 V C_L = 50 pF R_L = 1.6 kΩ ±5%	01	5	40	ns
Propagation delay time, high-to-low output from data to $\overline{EO}$	t_{PHL1}		01	10	75	ns
Propagation delay time, low-to-high output from $\overline{EI}$ to $\overline{GS}$	t_{PLH2}		01	8	45	ns
Propagation delay time, high-to-low output from $\overline{EI}$ to $\overline{GS}$	t_{PHL2}		01	8	53	ns
Propagation delay time, low-to-high output from $\overline{EI}$ to $\overline{EO}$	t_{PLH3}		01	8	45	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/	Device type	Limits Min	Limits Max	Unit
Propagation delay time, high-to-low output from $\overline{EI}$ to $\overline{EO}$	t_{PHL3}	V_{CC} = 5.0 V C_L = 50 pF R_L = 1.6 kΩ ±5%	01	8	61	ns
Propagation delay time, low-to-high output from $\overline{EI}$ to $\overline{A}$	t_{PLH4}		01	10	58	ns
Propagation delay time, high-to-low output from $\overline{EI}$ to $\overline{A}$	t_{PHL4}		01	8	51	ns
Propagation delay time, low-to-high output from data to $\overline{GS}$	t_{PLH5}		01	25	85	ns
Propagation delay time, high-to-low output from data to $\overline{GS}$	t_{PHL5}		01	8	61	ns
Propagation delay time, low-to-high output from data to $\overline{A}$	t_{PLH6}		01	8	61	ns
Propagation delay time, high-to-low output from data to $\overline{A}$	t_{PHL6}		01	8	61	ns

1/ Complete terminal conditions shall be as specified in table III.

2/ Not more than one output should be shorted at a time.

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TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)
	Class B devices
Interim electrical test parameters (pre burn-in) (method 5004)	1
Final electrical test parameters (method 5004)	1*,2,3,7, 9
Group A test requirements (method 5005)	1,2,3,9, 10,11
Group C end-point electrical test parameters (method 5005)	1,2,3
Group D end-point electrical test parameters (method 5005)	1,2,3
Additional electrical test parmeters for group C inspection	None

*PDA applies to subgroup 1 (see 4.2c).

3.7 Manufacturer eligibility. To be eligible to supply microcircuits to this specification, a manufacturer shall have manufacturer certification in accordance with MIL-M-38510 for at least one line. Not necessarily the line producing the device type described herein.

3.8 Certification. Certification in accordance with MIL-M-38510 is not required for this device.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameter tests prior to burn-in are optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) for class B devices shall be 10 percent based on failures from group A, subgroup 1 tests after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection is not required.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4). Generic test data (see 6.7) may be used to satisfy the requirements for groups C and D inspections. Quality conformance inspection shall be completed on the specific devices covered by this specification before they are shipped.

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4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II.

b. Subgroups 4, 5, 6, 7, and 8 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical test parameters shall be as specified in table II herein.

b. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A inspection.

c. Steady state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical test parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging of microcircuits shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for logistic support of existing equipment.

6.3 Ordering data. The contract or purchase order should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for packaging and packing.

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Device type 01

Cases E and F

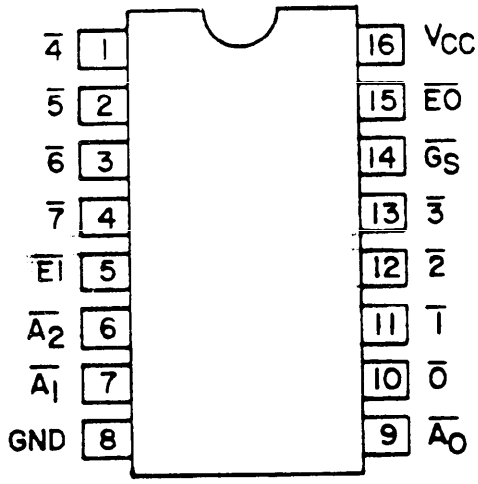


FIGURE 1. Terminal connections (top view).

Device type 01

Truth table (each gate)													
Input									Output				
$\overline{E1}$	$\overline{0}$	$\overline{1}$	$\overline{2}$	$\overline{3}$	$\overline{4}$	$\overline{5}$	$\overline{6}$	$\overline{7}$	$\overline{GS}$	$\overline{A_0}$	$\overline{A_1}$	$\overline{A2}$	$\overline{EO}$
H	X	X	X	X	X	X	X	X	H	H	H	H	H
L	H	H	H	H	H	H	H	H	H	H	H	H	L
L	X	X	X	X	X	X	X	L	L	L	L	L	H
L	X	X	X	X	X	X	L	H	L	H	L	L	H
L	X	X	X	X	X	L	H	H	L	L	H	L	H
L	X	X	X	X	L	H	H	H	L	H	H	L	H
L	X	X	X	L	H	H	H	H	L	L	L	H	H
L	X	X	L	H	H	H	H	H	L	H	L	H	H
L	X	L	H	H	H	H	H	H	L	L	H	H	H
L	L	H	H	H	H	H	H	H	L	H	H	H	H

NOTE:
H = high level (steady-state)
L = low level (steady-state)
X = irrelevant (any input, including transitions).

FIGURE 2. Truth table.

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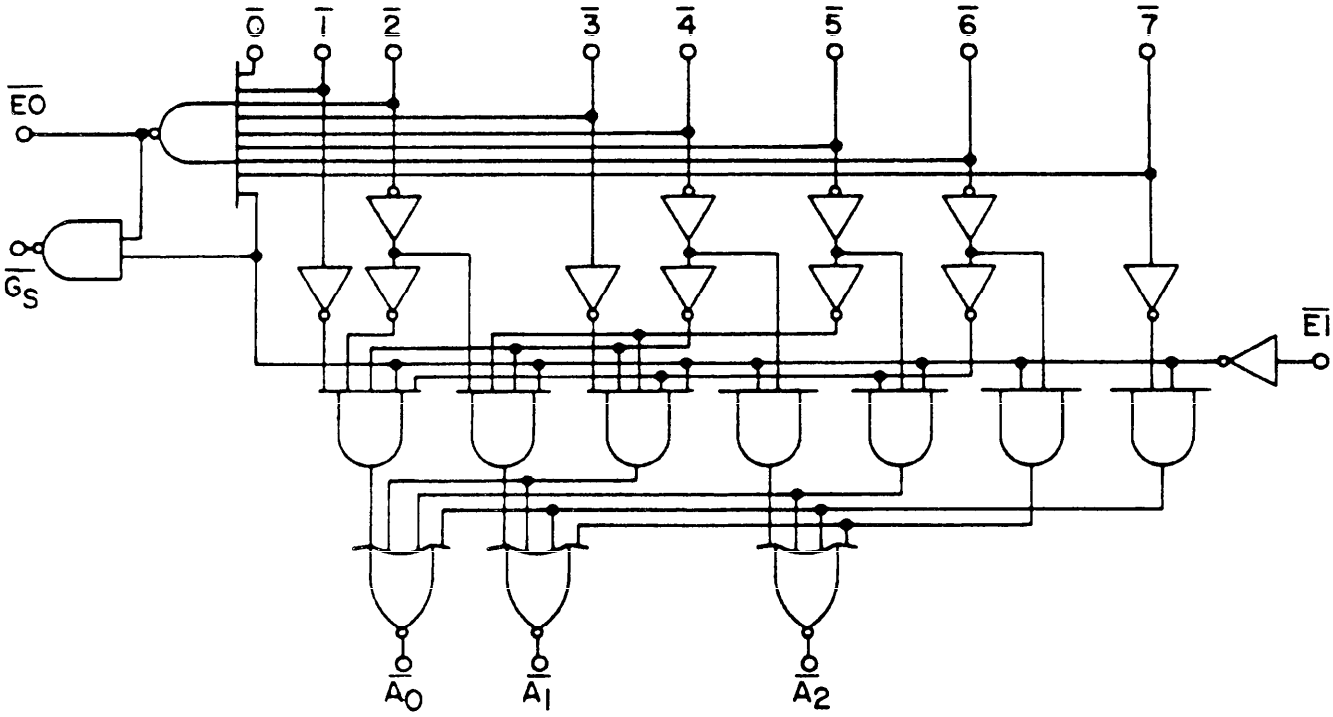
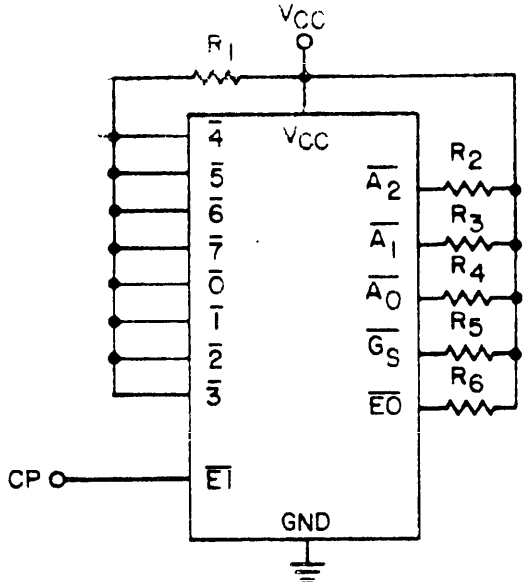


FIGURE 3. Logic diagram.

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of some future configuration. A demonstrated the hardware tested and per

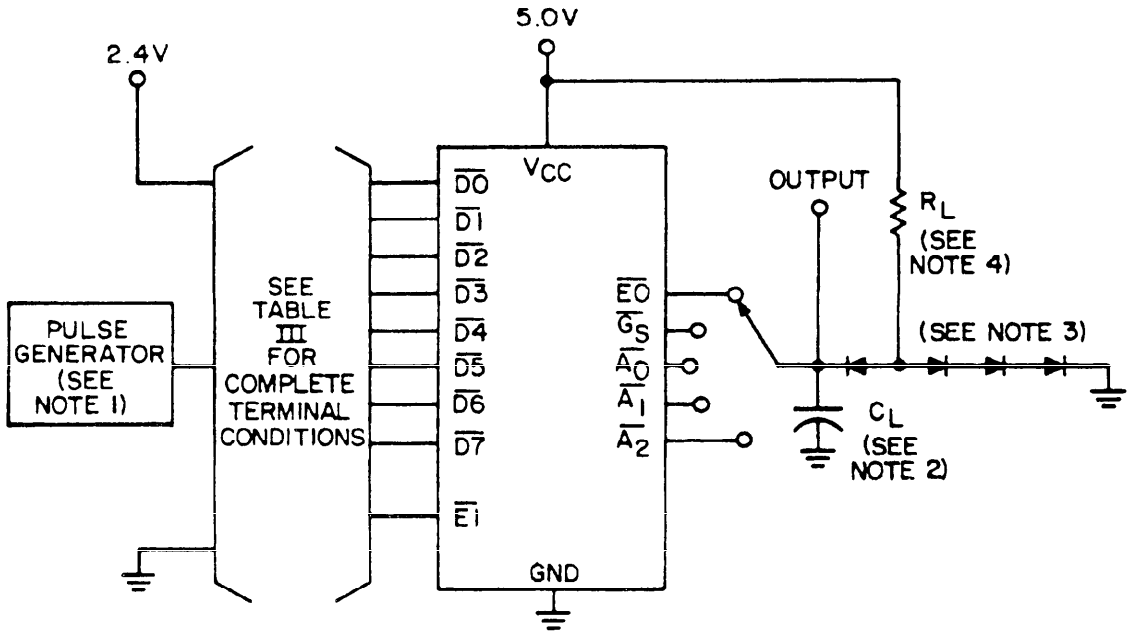
Device type 01



NOTES:
1. R_1 = 16 kΩ ±5%.
2. R_2 thru R_6 = 1.1 kΩ ±5%.
3. V_{CC} shall be of such a value as to provide 5.0 V minimum at device terminals.
4. CP = 1 MHz ±50% square wave; duty cycle = 50 ±15%; V_{IH} = 2.0 V minimum to 5.5 V maximum; V_{IL} = -0.5 V minimum to 0.7 V maximum.

FIGURE 4. Burn-in and steady state life test circuit.

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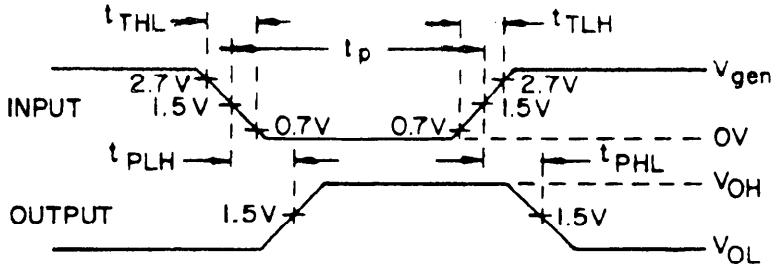


NOTES:
1. The pulse generator shall have the following characteristics: PRR = 1 MHz, t_p = 100 ns, $Z_{out} \approx 50\Omega$, $t_{TLH} = t_{THL} \leq 10$ ns, V_{gen} = 3.0 V.
2. C_L = 50 pF minimum including probe and jig capacitance.
3. All diodes are 1N3064 or equivalent.
4. R_L = 1.6 kΩ ±5%.

FIGURE 5. Switching time test circuit and waveforms for device type 01.

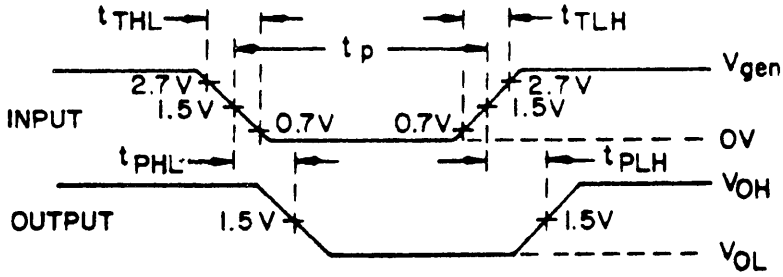
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Data input to enable output (t_{PLH1} and t_{PHL1})
Data input to data output (t_{PLH6} and t_{PHL6})



NOTE: Complete terminal conditions shall be as specified in table III.

Enable input to group signal (t_{PLH2} and t_{PHL2})
Enable input to enable output (t_{PLH3} and t_{PHL3})
Enable input to data output (t_{PLH4} and t_{PHL4})
Data input to group signal (t_{PLH5} and t_{PHL5})



NOTE: Complete terminal conditions shall be as specified in table III.

FIGURE 5. Switching time test circuit and waveforms for device type 01 - Continued.

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TABLE III. Group A inspection for device type 01.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open)

Subgroup	Symbol	MIL-STD-883 method	Case E & F Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limit Min	Test limit Max	Test limit Unit
1 T_C = 25°C	I_{OH}	3006	1	$\overline{1}$	5	$\overline{6}$	7	$\overline{E1}$	A2	$\overline{A1}$	GND	$\overline{A0}$	$\overline{0}$	$\overline{1}$	2	3	$\overline{GS}$	E0	V_{CC}	E0	3.4		V
			2					GND			GND		0.7 V	0.7 V				0.4mA	4.5 V				
			3																				
			4																				
			5	0.7 V																			
			6		0.7 V											0.7 V							
			7			0.7 V																	
			8				0.7 V																
			9					2.0 V												GS			
			10					2.0 V					GND				-0.4 mA			GS			
			11					GND				-0.4 mA		2.6 V			-0.4 mA			A0			
			12							-0.4mA					2.0 V					A1			
			13						0.4 mA									4.6 mA		A2			
			14	2.0 V				0.7 V					GND				4.8 mA			CS			
			15					GND		4.8mA		4.8mA		GND	GND					A			
			16						4.8mA						GND					A1			
			17																	A2			
			18																				
			19	GND 10mA																6			
			20																				
			21		-10 mA								-10 mA							E0			
			22			-10mA								-10 mA						0			
			23				-10 mA								-10 mA					1			
			24					-10 mA								-10 mA				2			
			25																	3			
			26																	4			
			27																	5			
			28					4.5 V											5.5 V	6			
		3010	29																	7			
	$\overline{E1}$		30	2.4 V																E1			
	I_{IH2}		31		2.4 V															0			
			32			2.4 V														1			
			33				2.4 V													2			
			34					2.4 V												3			
			35					4.5 V												4			
			36					4.5 V												5			
			37					4.5 V												6			
			38																	7			
	I_{IH3}		39	5.5 V																E1			
			40		5.5 V															0			
			41			5.5 V														1			
			42				5.5 V													2			
			43																	3			
			44																	4			
			45																	5			
			46																	6			

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open)

Subgroup	Symbol	MIL-STD-883 method	Case E & F Test No.	1 $\overline{4}$	2 $\overline{5}$	3 $\overline{6}$	4 $\overline{7}$	5 $\overline{E1}$	6 $\overline{A2}$	7 $\overline{A1}$	8 GND	9 $\overline{A0}$	10 $\overline{0}$	11 $\overline{1}$	12 $\overline{2}$	13 $\overline{3}$	14 $\overline{GS}$	15 $\overline{E0}$	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 T_C = 25°C	I_{IL1}	3009	47					GND			GND		0.3 V						5.5 V	$\overline{0}$	-176	-400	µA
	I_{IL2}		48	0.3 V	0.3 V	0.3 V	0.3 V	0.3 V						0.3 V	0.3 V	0.3 V				$\overline{4}$	-352	-800	
			49					GND												5			
			50					GND												6			
			51					GND												7			
			52																	$\overline{E1}$			
			53																	1			
			54																	2			
			55															GND		3			
	I_{OS}	3011	56					4.5 V									GND			$\overline{E0}$	-2.5	-25	mA
			57																	$\overline{GS}$			
			58									GND								$\overline{A0}$			
			59							GND										$\overline{A1}$			
			60						GND											$\overline{A2}$			
	I_{CC}	3005	61				GND	GND												V_{CC}		22	
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
9 T = 25°C	t_{PLH1}	3003 Fig 5	62	IN	IN	IN	IN	GND			GND		IN	IN	IN	IN		OUT	5.0 V	$\overline{0}$ - $\overline{E0}$	5	30	ns
			63																	$\overline{1}$ - $\overline{E0}$			
			64																	$\overline{2}$ - $\overline{E0}$			
			65																	$\overline{3}$ - $\overline{E0}$			
			66																	$\overline{4}$ - $\overline{E0}$			
			67																	$\overline{5}$ - $\overline{E0}$			
			68																	$\overline{6}$ - $\overline{E0}$			
			69																	$\overline{7}$ - $\overline{E0}$			
	t_{PHL1}		70	IN	IN	IN	IN	GND			GND		IN	IN	IN	IN		OUT	5.0 V	$\overline{0}$ - $\overline{E0}$	15	65	
			71																	$\overline{1}$ - $\overline{E0}$			
			72																	$\overline{2}$ - $\overline{E0}$			
			73																	$\overline{3}$ - $\overline{E0}$			
			74																	$\overline{4}$ - $\overline{E0}$			
			75																	$\overline{5}$ - $\overline{E0}$			
			76																	$\overline{6}$ - $\overline{E0}$			
			77																	$\overline{7}$ - $\overline{E0}$			
	t_{PLH2}		78					IN					GND				OUT			$\overline{E1}$ - $\overline{GS}$	8	35	
	t_{PHL2}		79										GND				OUT			$\overline{E1}$ - $\overline{GS}$	10	43	
	t_{PLH3}		80															OUT		$\overline{E1}$ - $\overline{E0}$	8	35	
	t_{PHL3}		81															OUT		$\overline{E1}$ - $\overline{E0}$	10	51	
	t_{PLH4}		82				GND					OUT								$\overline{E1}$ - $\overline{A0}$	10	48	
	t_{PLH4}		83							OUT										$\overline{E1}$ - $\overline{A1}$	10	48	
	t_{PLH4}		84						OUT											$\overline{E1}$ - $\overline{A2}$	10	48	
	t_{PHL4}		85									OUT								$\overline{E1}$ - $\overline{A0}$	8	41	
	t_{PHL4}		86							OUT										$\overline{E1}$ - $\overline{A1}$	8	41	
	t_{PHL4}		87						OUT											$\overline{E1}$ - $\overline{A2}$	8	41	

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high $\geq$ 2.0 V or low $\leq$ 0.7 V or open)

Subgroup	Symbol	MIL-STD-883 method	Case E & F Test No.	1 $\overline{4}$	2 $\overline{5}$	3 $\overline{6}$	4 $\overline{7}$	5 $\overline{E1}$	6 $\overline{A2}$	7 $\overline{A1}$	8 GND	9 $\overline{A0}$	10 $\overline{0}$	11 $\overline{1}$	12 $\overline{2}$	13 $\overline{3}$	14 $\overline{GS}$	15 $\overline{E0}$	16 V_{CC}	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
9 T_C 25°C	t_{PLH5} Fig 5	3003	88					GND			GND		IN	IN			OUT		5.0 V	$\overline{0}$ - $\overline{GS}$	30	75	ns
			89												IN					$\overline{1}$ - $\overline{GS}$			
			90													IN				$\overline{2}$ - $\overline{GS}$			
			91																	$\overline{3}$ - $\overline{GS}$			
			92	IN																$\overline{4}$ - $\overline{GS}$			
			93		IN															$\overline{5}$ - $\overline{GS}$			
			94			IN														$\overline{6}$ - $\overline{GS}$			
			95				IN													$\overline{7}$ - $\overline{GS}$			
	t_{PHL5} Fig 5		96										IN							$\overline{0}$ - $\overline{GS}$	10	51	
			97											IN						$\overline{1}$ - $\overline{GS}$			
			98												IN					$\overline{2}$ - $\overline{GS}$			
			99													IN				$\overline{3}$ - $\overline{GS}$			
			100	IN																4 - $\overline{GS}$			
			101		IN															5 - $\overline{GS}$			
			102			IN														6 - $\overline{GS}$			
			103				IN													7 - $\overline{GS}$			
	t_{PLH6} Fig 5		104									OUT		IN						$\overline{1}$ - A0			
			105									OUT			IN					$\overline{2}$ - $\overline{A0}$			
			106							OUT					IN					$\overline{2}$ - $\overline{A1}$			
			107									OUT				IN				$\overline{3}$ - $\overline{A0}$			
			108							OUT						IN				$\overline{3}$ - $\overline{A1}$			
			109	IN								OUT				GND				$\overline{4}$ - $\overline{A0}$			
			110	IN						OUT						GND				$\overline{4}$ - $\overline{A1}$			
			111	IN					OUT							GND				$\overline{4}$ - $\overline{A2}$			
			112		IN							OUT			GND					5 - $\overline{A0}$			
			113		IN					OUT					GND					$\overline{5}$ - $\overline{A1}$			
			114		IN				OUT						GND					$\overline{5}$ - $\overline{A2}$			
			115			IN						OUT		GND						$\overline{6}$ - $\overline{A0}$			
			116			IN				OUT				GND						$\overline{6}$ - $\overline{A1}$			
			117			IN			OUT					GND						$\overline{6}$ - $\overline{A2}$			
			118				IN					OUT								$\overline{7}$ - $\overline{A0}$			
			119				IN			OUT										$\overline{7}$ - $\overline{A1}$			
			120				IN		OUT											$\overline{7}$ - $\overline{A2}$			
	t_{PHL6} Fig 5		121									OUT		IN						$\overline{1}$ - $\overline{A0}$			
			122									OUT			IN					$\overline{2}$ - $\overline{A0}$			
			123							OUT					IN					$\overline{2}$ - $\overline{A1}$			
			124									OUT				IN				$\overline{3}$ - $\overline{A0}$			
			125							OUT						IN				$\overline{3}$ - $\overline{A1}$			
			126	IN								OUT				GND				$\overline{4}$ - $\overline{A0}$			
			127	IN						OUT						GND				$\overline{4}$ - $\overline{A1}$			
			128	IN					OUT							GND				$\overline{4}$ - $\overline{A2}$			
			129		IN							OUT			GND					5 - $\overline{A0}$			
			130		IN					OUT					GND					$\overline{5}$ - $\overline{A1}$			
			131		IN				OUT						GND					$\overline{5}$ - $\overline{A2}$			
			132			IN						OUT		GND						$\overline{6}$ - $\overline{A0}$			
			133			IN				OUT				GND						$\overline{6}$ - $\overline{A1}$			
			134			IN			OUT					GND						$\overline{6}$ - $\overline{A2}$			
			135				IN					OUT								$\overline{7}$ - $\overline{A0}$			
			136				IN			OUT										$\overline{7}$ - $\overline{A1}$			
			137				IN		OUT											$\overline{7}$ - $\overline{A2}$			

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open)

Subgroup	Symbol	MIL-STD-883 method	Case E & F Test No.	1 $\overline{4}$	2 $\overline{5}$	3 $\overline{6}$	4 $\overline{7}$	5 $\overline{E1}$	6 $\overline{A2}$	7 $\overline{A1}$	8 GND	9 $\overline{A0}$	10 $\overline{0}$	11 $\overline{1}$	12 $\overline{2}$	13 $\overline{3}$	14 $\overline{GS}$	15 $\overline{E0}$	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
10 T_C=125°C	t_{PLH1} Fig 5	3003	138	IN	IN	IN	IN	GND			GND		IN	IN	IN	IN		OUT	5.0 V	$\overline{0}$ - $\overline{E0}$	5	40	ns
			139																	$\overline{1}$ - $\overline{E0}$			
			140																	$\overline{2}$ - $\overline{E0}$			
			141																	$\overline{3}$ - $\overline{E0}$			
			142																	$\overline{4}$ - $\overline{E0}$			
			143																	$\overline{5}$ - $\overline{E0}$			
			144																	$\overline{6}$ - $\overline{E0}$			
			145																	$\overline{7}$ - $\overline{E0}$			
	t_{PHL1} Fig 5		146	IN	IN	IN	IN						IN	IN	IN	IN				$\overline{0}$ - $\overline{E0}$	10	75	
			147																	$\overline{1}$ - $\overline{E0}$			
			148																	$\overline{2}$ - $\overline{E0}$			
			149																	$\overline{3}$ - $\overline{E0}$			
			150																	$\overline{4}$ - $\overline{E0}$			
			151																	$\overline{5}$ - $\overline{E0}$			
			152																	$\overline{6}$ - $\overline{E0}$			
			153																	$\overline{7}$ - $\overline{E0}$			
	t_{PLH2} t_{PHL2}		154					IN					GND				OUT			$\overline{E1}$ - $\overline{GS}$	8	45	
			155										GND				OUT			$\overline{E1}$ - $\overline{GS}$	8	53	
	t_{PLH3} t_{PHL3}		156															OUT		$\overline{E1}$ - $\overline{E0}$	8	45	
			157															OUT		$\overline{E1}$ - $\overline{E0}$	8	61	
	t_{PLH4}		158				GND			OUT		OUT								$\overline{E1}$ - $\overline{A0}$	10	58	
	t_{PLH4}		159																	$\overline{E1}$ - $\overline{A1}$	10	58	
	t_{PLH4}		160						OUT											$\overline{E1}$ - $\overline{A2}$	10	58	
	t_{PHL4}		161																	$\overline{E1}$ - $\overline{A0}$	8	51	
	t_{PHL4}		162																	$\overline{E1}$ - $\overline{A1}$	8	51	
	t_{PHL4}		163						OUT											$\overline{E1}$ - $\overline{A2}$	8	51	
	t_{PLH5} Fig 5		164	IN	IN	IN	IN	GND					IN	IN	IN	IN	OUT			$\overline{0}$ - $\overline{GS}$	25	85	
			165																	$\overline{1}$ - $\overline{GS}$			
			166																	$\overline{2}$ - $\overline{GS}$			
			167																	$\overline{3}$ - $\overline{GS}$			
			168																	$\overline{4}$ - $\overline{GS}$			
			169																	$\overline{5}$ - $\overline{GS}$			
			170																	$\overline{6}$ - $\overline{GS}$			
			171																	$\overline{7}$ - $\overline{GS}$			
	t_{PHL5} Fig 5		172	IN	IN	IN	IN						IN	IN	IN	IN				$\overline{0}$ - $\overline{GS}$	8	61	
			173																	$\overline{1}$ - $\overline{GS}$			
			174																	$\overline{2}$ - $\overline{GS}$			
			175																	$\overline{3}$ - $\overline{GS}$			
			176																	$\overline{4}$ - $\overline{GS}$			
			177																	$\overline{5}$ - $\overline{GS}$			
			178																	$\overline{6}$ - $\overline{GS}$			
			179																	$\overline{7}$ - $\overline{GS}$			

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high $\geq$ 2.0 V or low $\leq$ 0.7 V or open)

Subgroup	Symbol	MIL-STD-883 method	Case E & F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No	$\overline{4}$	$\overline{5}$	$\overline{6}$	$\overline{7}$	$\overline{E1}$	$\overline{A2}$	$\overline{A1}$	GND	$\overline{A0}$	$\overline{0}$	$\overline{1}$	$\overline{2}$	$\overline{3}$	$\overline{GS}$	$\overline{E0}$	V_{CC}		Min	Max	Unit
10 T_C = 125°C	t_{PLH6} Fig 5	3003	180					GND			GND	OUT		IN					5.0 V	$\overline{1}$ - $\overline{A0}$	8	61	ns
			181									OUT			IN					$\overline{2}$ - $\overline{A0}$			
			182							OUT					IN					$\overline{2}$ - $\overline{A1}$			
			183									OUT				IN				$\overline{3}$ - $\overline{A0}$			
			184							OUT						IN				$\overline{3}$ - $\overline{A1}$			
			185	IN								OUT				GND				$\overline{4}$ - $\overline{A0}$			
			186	IN						OUT						GND				$\overline{4}$ - $\overline{A1}$			
			187	IN					OUT							GND				$\overline{4}$ - $\overline{A2}$			
			188		IN							OUT			GND					$\overline{5}$ - $\overline{A0}$			
			189		IN					OUT					GND					$\overline{5}$ - $\overline{A1}$			
			190		IN				OUT						GND					$\overline{5}$ - $\overline{A2}$			
			191			IN						OUT		GND						$\overline{6}$ - $\overline{A0}$			
			192			IN				OUT				GND						$\overline{6}$ - $\overline{A1}$			
			193			IN			OUT					GND						$\overline{6}$ - $\overline{A2}$			
			194				IN					OUT								$\overline{7}$ - $\overline{A0}$			
			195				IN			OUT										$\overline{7}$ - $\overline{A1}$			
			196				IN		OUT											$\overline{7}$ - $\overline{A2}$			
	t_{PHL6} Fig 5		197									OUT		IN						$\overline{1}$ - $\overline{A0}$			
			198									OUT			IN					$\overline{2}$ - $\overline{A0}$			
			199							OUT					IN					$\overline{2}$ - $\overline{A1}$			
			200									OUT				IN				$\overline{3}$ - $\overline{A0}$			
			201							OUT						IN				$\overline{3}$ - $\overline{A1}$			
			202	IN								OUT				GND				$\overline{4}$ - $\overline{A0}$			
			203	IN						OUT						GND				$\overline{4}$ - $\overline{A1}$			
			204	IN					OUT							GND				$\overline{4}$ - $\overline{A2}$			
			205		IN							OUT			GND					$\overline{5}$ - $\overline{A0}$			
			206		IN					OUT					GND					$\overline{5}$ - $\overline{A1}$			
			207		IN				OUT						GND					$\overline{5}$ - $\overline{A2}$			
			208			IN						OUT		GND						$\overline{6}$ - $\overline{A0}$			
			209			IN				OUT				GND						$\overline{6}$ - $\overline{A1}$			
			210			IN			OUT					GND						$\overline{6}$ - $\overline{A2}$			
			211				IN					OUT								$\overline{7}$ - $\overline{A0}$			
			212				IN			OUT										$\overline{7}$ - $\overline{A1}$			
			213				IN		OUT											$\overline{7}$ - $\overline{A2}$			
11	Same tests, terminal conditions and limits as subgroup 10, except T_C = -55°C.																						

MIL-M-38510/4 J19

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f. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Electrical ground (common terminal)
I_{IN}	Current flowing into an input terminal
V_{IN}	Voltage level at an input terminal
T_C	Case temperature

6.5 Logistic support. Lead materials and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Generic test data. Generic test data may be used to satisfy the requirements of 4.4.3. Group C generic test data shall be on date codes not more than one year old and on a die in the same microcircuit group (see appendix E of MIL-M-38510) with the same material, design and process and from the same plant as the die represented. Group D (see 4.4.3) generic data shal be on date codes no more than one year old and on the same package type (see terms, definitions, and sybmols of MIL-M-38510) and from the same package represented. The vendor is required to retain the generic data for a period of not less than 36 months from the date of shipment.

6.7 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	93L18

6.8 Ordering guidance. Since the qualification and certification requirements have been removed from the specification, orders may be placed immediately.

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
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Navy - OS, SH
Air Force - 11, 19, 85, 99
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User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/43B	Microcircuits, Digital, TTL, Low Power, Priority Encoders, Monolithic Silicon

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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